



EXAM PAPERS PRACTICE

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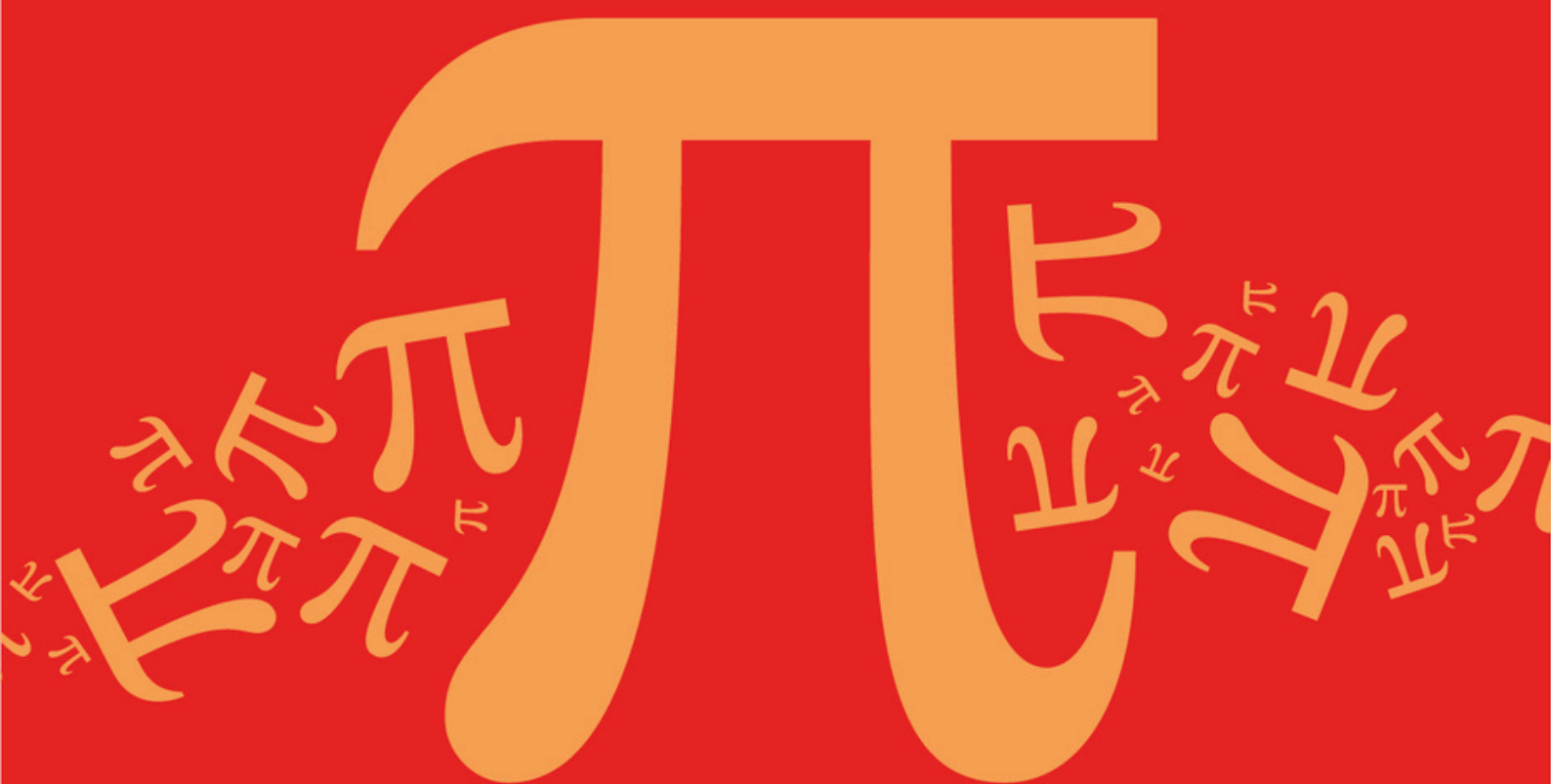
Practice questions created by actual examiners and assessment experts

Detailed mark scheme

Suitable for all boards

Designed to test your ability and thoroughly prepare you

AQA A Level Further Mathematics (7367)



Topic

D: Further Algebra and Functions

Sub topic

D12: Graphs of Rational Functions - Asymptotes

Topic Questions

AQA A Level Further Mathematics (7367)

Question Paper

Topic

D: Further Algebra and Functions

Sub topic

D12: Graphs of Rational Functions - Asymptotes

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Name : _____

Date : _____

Time : _____

Total Marks Available : _____

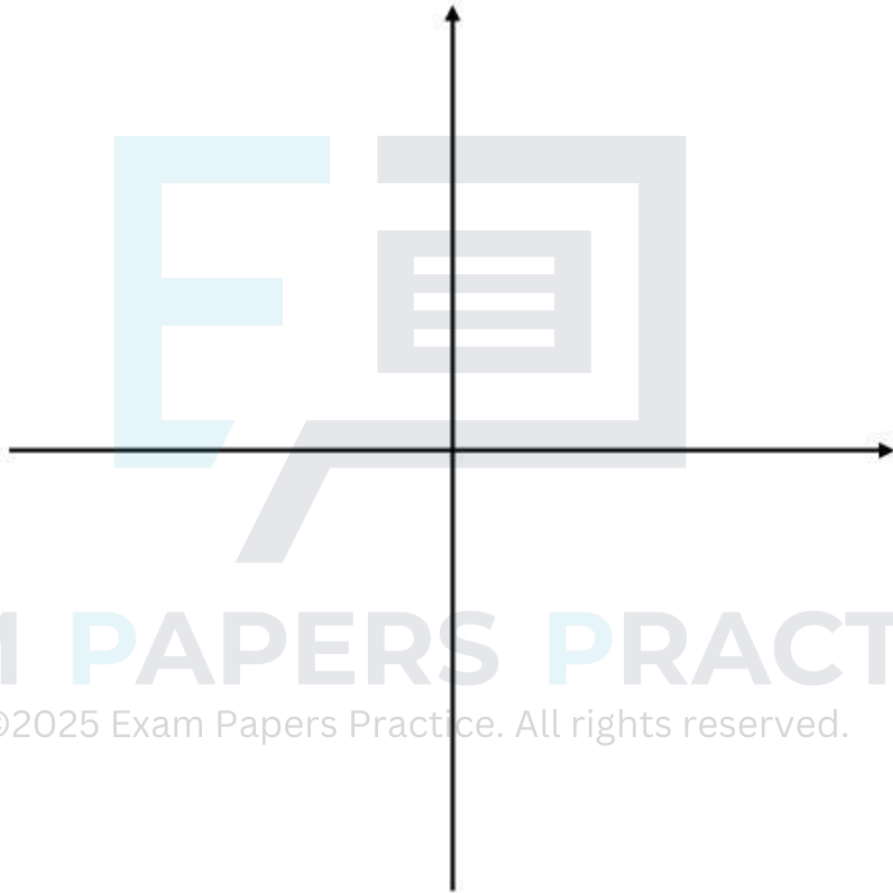
Total Marks Received : _____

Q1.

A curve has equation

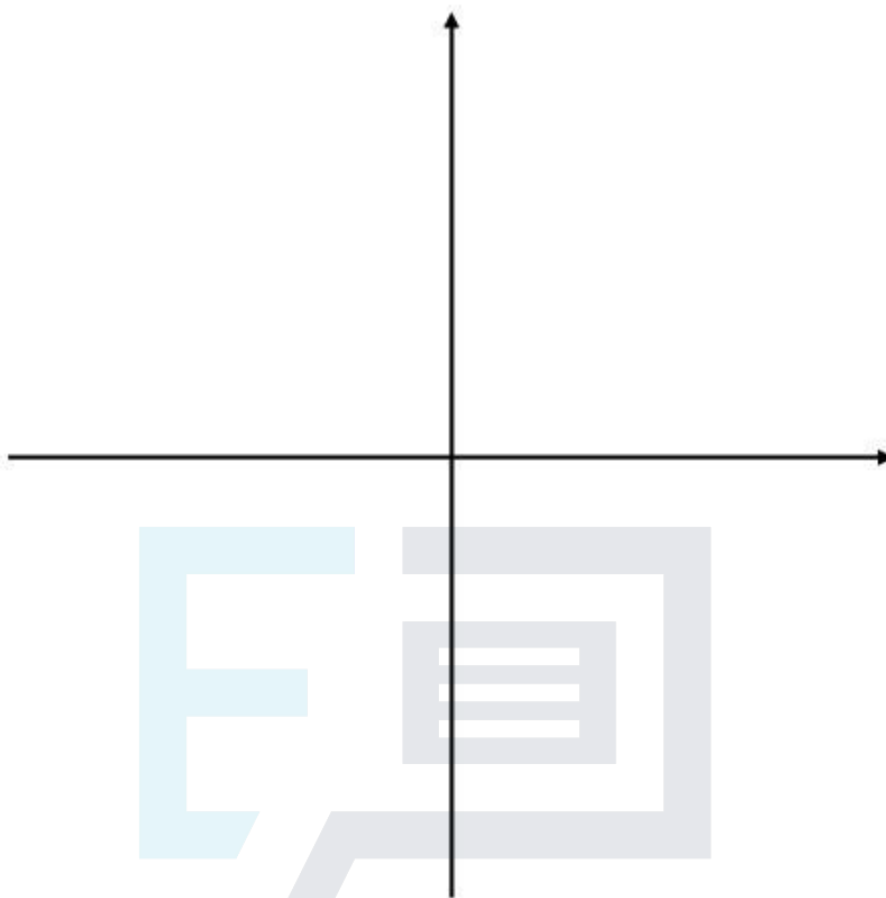
$$y = \frac{5-4x}{1+x}$$

(a) Sketch the curve.



(4)

(b) Hence sketch the graph of $y = \left| \frac{5-4x}{1+x} \right|$.



(1)
(Total 5 marks)

Q2.

A curve has equation

$$y = \frac{x}{x-1}$$

- (a) Find the equations of the asymptotes of this curve.

(2)

- (b) Given that the line $y = -4x + c$ intersects the curve, show that the x -coordinates of the points of intersection must satisfy the equation

$$4x^2 - (c + 3)x + c = 0$$

(3)

- (c) It is given that the line $y = -4x + c$ is a tangent to the curve.

- (i) Find the two possible values of c .

(No credit will be given for methods involving differentiation.)

(3)

- (ii) For each of the two values found in part (c)(i), find the coordinates of the point where the line touches the curve.

(4)
(Total 12 marks)

Q3.

A curve has equation $y = \frac{1}{x^2 - 4}$.

(a) (i) Write down the equations of the three asymptotes of the curve. (3)

(ii) Sketch the curve, showing the coordinates of any points of intersection with the coordinate axes. (4)

(b) Hence, or otherwise, solve the inequality

$$\frac{1}{x^2 - 4} < -2$$

(3)
(Total 10 marks)

Q4.

(a) (i) Write down the equations of the two asymptotes of the curve $y = \frac{1}{x - 3}$. (2)

(ii) Sketch the curve $y = \frac{1}{x - 3}$, showing the coordinates of any points of intersection with the coordinate axes. (2)

(iii) On the same axes, again showing the coordinates of any points of intersection with the coordinate axes, sketch the line $y = 2x - 5$. (1)

(b) (i) Solve the equation

$$\frac{1}{x - 3} = 2x - 5$$

(3)

(ii) Find the solution of the inequality

$$\frac{1}{x - 3} < 2x - 5$$

(2)
(Total 10 marks)

Q5.

A curve has equation

$$y = \frac{3x-1}{x+2}$$

- (a) Write down the equations of the two asymptotes to the curve. (2)
- (b) Sketch the curve, indicating the coordinates of the points where the curve intersects the coordinate axes. (5)
- (c) Hence, or otherwise, solve the inequality

$$0 < \frac{3x-1}{x+2} < 3$$

(2)

(Total 9 marks)

Q6.

A curve has equation

$$y = \frac{6x}{x-1}$$

- (a) Write down the equations of the two asymptotes to the curve. (2)
- (b) Sketch the curve and the two asymptotes. (4)
- (c) Solve the inequality

$$\frac{6x}{x-1} < 3$$

(4)

(Total 10 marks)